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RPMC Training

Why a good investment needs a good portfolio

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Rotman School of Management
UNIVERSITY OF TORONTO

You like the company. How much should you own?

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A business judgment does not yet tell you the right portfolio weight

THE BUSINESS

Will it do well?

Customers, profits, cash generation, and competition.

THE PRICE

What is already expected?

A strong business can be too expensive.

THE POSITION

What if we are wrong?

The same investment creates different losses at different sizes.

If the shares fall 30%: a 10% portfolio weight loses 3% of the account; a 100% weight loses 30%.

What are we actually buying?

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Stock, ETF, and bond describe different claims on investments

STOCK

A share of a business

Returns come from changes in price and distributions such as dividends.

ETF

A tradable fund

What it owns determines the exposure: stocks, bonds, or something else.

BOND

A claim on payments

An issuer promises interest and principal; price and repayment are uncertain.

An ETF is a wrapper. A bond ETF still has bond risks; a sector ETF can still be concentrated.

First, count what the investment actually earns

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Return includes both price change and income

PURCHASE PRICE

\$100

ENDING PRICE

\$106

DIVIDEND RECEIVED

\$2

$$R = \frac{P_1 - P_0 + \text{Dividend}}{P_0}$$

$$(\$106 - \$100 + \$2) \div \$100 = 8\%$$

R = holding-period return; P_0 = purchase price; P_1 = ending price. Ignore costs and taxes in this example.

Why diversification can reduce risk

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Spread company-specific uncertainty; recognize the market risks that remain

COMPANY-SPECIFIC RISK

A setback in one business

Different businesses face different individual shocks. Spreading holdings can reduce dependence on any one outcome.

SYSTEMATIC RISK

A shock shared across the market

A broad equity portfolio can still lose when market-wide conditions deteriorate.

Diversification depends on position weights and how returns move together, not simply the number of holdings.

What makes two holdings different sources of risk?

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Use economic exposures and observed co-movement together

ECONOMIC DRIVERS

Look beyond the ticker

Different companies can depend on the same customers, funding conditions, or economic cycle.

CORRELATION

Measure co-movement

Correlation summarizes the linear relationship between return fluctuations.

CHANGING CONDITIONS

Test the relationship

Historical relationships may weaken or change when market-wide shocks dominate.

Correlation ranges from -1 to $+1$. Zero correlation does not rule out simultaneous losses.

Which experience of risk are we exposing?

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Fluctuation, loss from a peak, and the ability to trade

VOLATILITY

How much do returns fluctuate?

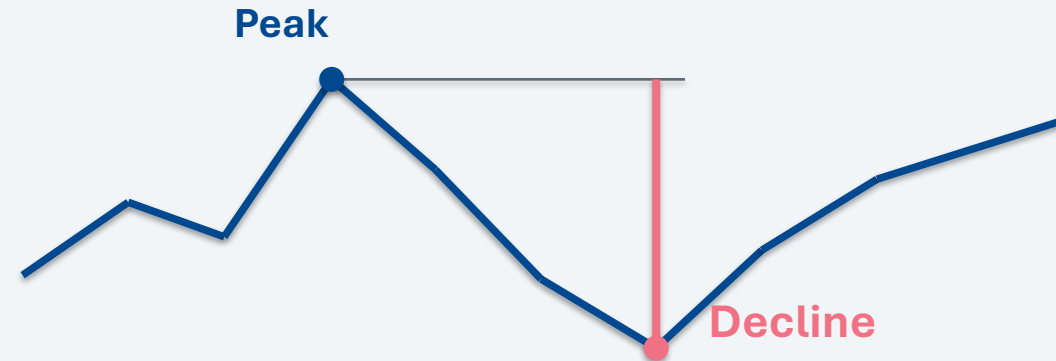
Standard deviation measures dispersion around the average.

LIQUIDITY

Can we trade when we need to?

Time, available quantity, and the price concession all matter.

DRAWDOWN: LOSS FROM A PREVIOUS PEAK



Schematic account path; no market data.

$$\text{Drawdown} = \frac{\text{Peak value} - \text{Current value}}{\text{Peak value}}$$

Use volatility to study co-movement; use drawdown and liquidity to examine the path an investor must withstand.

A holding's risk depends on the company it keeps

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The variance model connects position size, individual risk, and co-movement

TWO-ASSET PORTFOLIO VARIANCE

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

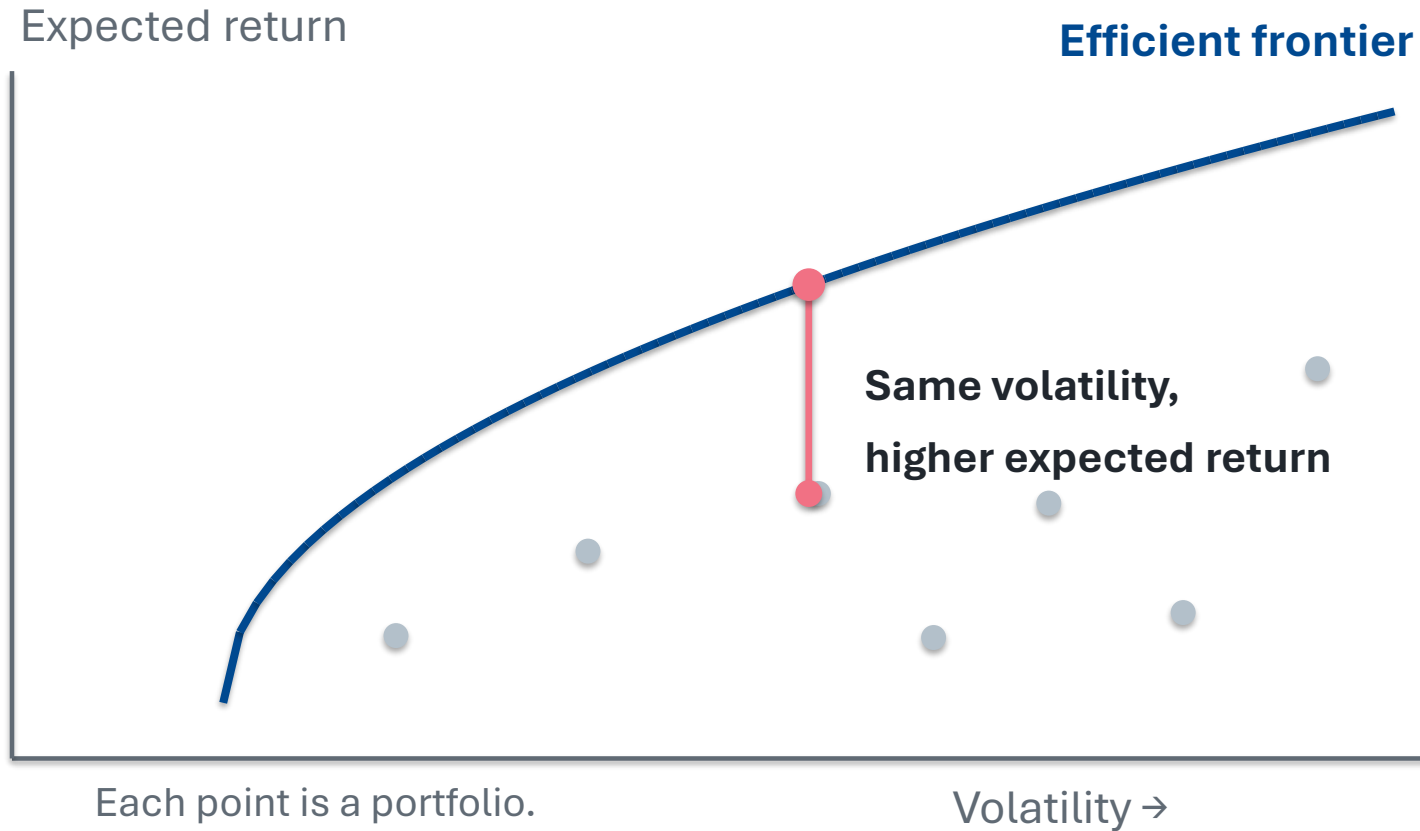
w = portfolio weight; σ = return volatility; ρ_{12} = correlation. The first two terms measure weighted own risk; the last captures co-movement.

Variance is volatility squared. Covariance = $\rho_{12} \times \sigma_1 \times \sigma_2$. Use the same return period for both assets.

Which portfolios make better use of risk?

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Compare expected return and volatility for the entire combination



TWO COMPARISONS

**More expected return
for the same risk.**

**Less risk for the same
expected return.**

Conceptual diagram. The feasible set depends on the assets, constraints, and return and covariance estimates.

Why should an investment earn a return?

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Separate compensation for exposure from an active investment judgment

RISK PREMIUM

Bear an exposure

Expected compensation for accepting uncertainty, including losses in difficult conditions.

ACTIVE JUDGMENT

Find a gap in the price

Evidence that future outcomes may differ from what the current price reflects.

A position can contain both market exposure and an active thesis. Understand the contribution of each.

How much of the movement comes with the market?

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Beta measures market sensitivity; volatility measures total fluctuation

BETA FROM MATCHED RETURN OBSERVATIONS

$$\beta_i = \frac{\text{Cov}(R_i - R_f, R_m - R_f)}{\text{Var}(R_m - R_f)}$$

R_i = investment return; R_m = market return; R_f = risk-free return. Cov measures joint variation; Var measures the market's own variation.

Beta depends on the market proxy, return frequency, currency, and estimation period.

What return compensates for market risk in CAPM?

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Expected return combines the risk-free rate and compensation for market exposure

CAPITAL ASSET PRICING MODEL

$$E[R_i] = R_f + \beta_i(E[R_m] - R_f)$$

$$\text{e.g., } 10\% = 3\% + 1.4 \times 5\%$$

$E[R_i]$ = expected investment return; R_f = risk-free rate; β_i = market beta. $E[R_m] - R_f$ is the expected market risk premium.

CAPM links expected return to market risk under its equilibrium assumptions; realized returns can differ substantially.

How much excess return per unit of volatility?

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StockTrak reports an annualized Sharpe Ratio in session rankings

$$\text{Annualized Sharpe} = \frac{\text{Average daily portfolio return} - \text{Daily risk-free rate}}{\text{Standard deviation of daily portfolio returns}} \times \sqrt{252}$$

AVERAGE DAILY RETURN

Exclude weekends

Calculate the average of daily portfolio returns, excluding weekend observations.

DAILY RISK-FREE RATE

Cash-interest reference

By default, use “interest earned on cash”:
3% annual interest for most sessions.

DAILY STANDARD DEVIATION

Exclude weekends

Divide by the standard deviation of daily portfolio returns, excluding weekends.

Where to see your Sharpe ranking

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Portfolio Simulation → My Portfolio → Portfolio Ranking

Portfolio Simulation



My Portfolio



Portfolio Ranking



View: Sharpe Ratio Ranking

RETURN DATES

Exclude weekends from the daily returns.

CALCULATION START

Sharpe begins five trading days after the first trade.

TEAM REVIEW

Track the metric alongside exposures and drawdown.

Historical data: Graph My Portfolio → export historical performance. Exported history includes weekend rows.

A good company can still be a disappointing investment **Rotman**

The opportunity depends on the price and the expectations inside it

YOUR VIEW

Profits will grow

What do customers, costs, and investment imply for future cash generation?

MARKET EXPECTATIONS

Growth may be priced in

What future performance would justify today's valuation?

THE OPPORTUNITY

The difference matters

What could happen that the current price does not adequately reflect?

“The business is strong” is a starting observation. Explain why its price offers an attractive return.

Build a strategy from belief to position

Investment characteristics, research methods, and position structures can be combined

Layer	The question	Examples
Investment belief	Why might this opportunity earn a return?	Risk premium; underappreciated value or quality; price persistence.
Research method	How will we form and test the view?	Company analysis; macro research; systematic signals and testing.
Position structure	Which exposure expresses the view?	Long-only; industry long–short; a currency or market hedge.

A systematic strategy can use fundamental data. Long–short describes the position structure, not its source of return.

Canada: growth and inflation send different signals

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Economic information available on October 7, 2026

POLICY RATE

2.25%

Held on September 2

Inflation upside risks and uncertain growth complicate the next move.

INFLATION · AUGUST

3.0%

Headline CPI, year over year

CPI-trim 1.9%; CPI-median 2.0%. Headline and underlying measures differ.

LABOUR · AUGUST

6.4%

Unemployment rate

Employment fell 42,000 after gains from April through July.

REAL GDP · MONTHLY

0.0%

July growth

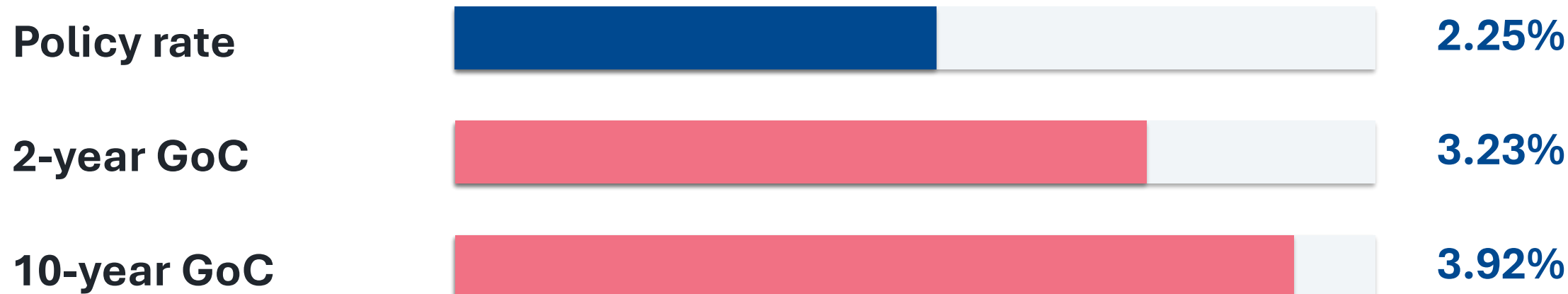
August advance estimate: +0.2%. Preliminary, not a confirmed rebound.

Read activity and inflation together, then trace the effect on cash flows, financing costs, and required returns.

The policy rate and the bond yield

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Canadian market observations on October 6, 2026



When required market yields rise, existing fixed payments lose value.

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \times \Delta y$$

D_{mod} = modified duration

Δy = yield change in decimal units

Duration measures rate sensitivity. A corporate bond also carries credit-spread risk; yield changes need not follow the policy rate.

Which surprise would change the investment case?

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Conditional scenarios; compare new evidence with what prices already imply

New evidence	Government bonds	Credit and equities	Currency question
Demand weakens; inflation broadening fades	Lower required yields could help duration.	Default concerns can offset falling base yields.	Does CAD weaken relative to the USD?
Energy / tariff costs persist despite soft growth	Higher inflation compensation can hurt duration.	Input costs squeeze firms without pricing power.	Oil support for CAD can conflict with trade risk.
Growth improves without wider inflation	Less easing than priced can lift yields.	Earnings and credit quality can improve.	Compare relative growth, rates, and risk appetite.

Compare new information with expectations already reflected in prices. Match the exposure to the country and risk being studied.

The asset can rise while the currency moves against you

CAD RETURN FROM AN UNHEDGED USD ASSET

$$R_{CAD} = (1 + R_{USD})(1 + R_{FX}) - 1$$

R_{USD} = asset return in USD; R_{FX} = the USD's return measured in CAD. Both legs affect the Canadian investor.

Trading currency and underlying currency exposure can differ. Size a Forex position by its economic notional.

What if our view is about the better retailer?

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Long the stronger business; short the weaker one — at prices that support the comparison

Evidence	Retailer A: buy	Retailer B: short
Customers and margins	Loyal customers; resilient margins.	Discounting to maintain demand.
Inventory and finances	Controlled stock; reliable cash generation.	Excess stock; debt pressure.
What the price misses	Strength is underappreciated.	Recovery is priced too optimistically.

Fictional companies in the same industry. The thesis requires a pricing gap, not just a quality difference.

The pair expresses a relative-performance view

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Equal starting dollar amounts: long Retailer A, short Retailer B

$$\text{Price P\&L} = N \times (R_A - R_B)$$

Market direction	If A outperforms B	Effect on the pair
Both rise	A rises more	Long gain exceeds short loss.
Both fall	A falls less	Short gain exceeds long loss.
A underperforms B	The relative view fails	The pair loses before costs.

N = equal starting position value; RA and RB = price returns in the same currency. Income, financing, borrow and trading costs also matter.

Gross and net exposure describe the position's scale

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Gross exposure measures how much is open; net exposure measures the dollar difference

GROSS: HOW MUCH IS OPEN?

$$\text{Gross exposure} = \frac{\text{Long value} + \text{Short value}}{\text{Account equity}}$$

Long value = market value of shares owned; short value = positive market value of shares owed.

NET: WHICH DIRECTION?

$$\text{Net exposure} = \frac{\text{Long value} - \text{Short value}}{\text{Account equity}}$$

Account equity = net portfolio value. Equal long and short amounts give zero net exposure for this pair.

For a CAD 1m account: CAD 150k long + CAD 150k short gives 30% gross and 0% net exposure.

Hedge the market exposure in the retailer pair

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Choose the exposure to offset before choosing the hedge ratio

MEASURE THE PAIR'S MARKET EXPOSURE

$$\beta_{\text{pair}} = \text{Long weight} \times \beta_A - \text{Short weight} \times \beta_B$$

Assume $\beta_A = 1.2$ and $\beta_B = 0.8$. Equal 15% weights: $0.15 \times 1.2 - 0.15 \times 0.8 = +0.06$.

CHOOSE THE SHORT TO MATCH THE LONG

$$\text{Short value} = \text{Long value} \times \frac{\beta_A}{\beta_B}$$

\$120k long A requires $\$120k \times 1.2 / 0.8 = \$180k$ short B. On \$1m equity: $0.12 \times 1.2 - 0.18 \times 0.8 = 0$.

Both weights are positive position magnitudes; the short is subtracted. New pair: gross 30%, net -6%, estimated beta 0.

A sound comparison still has to survive the trade

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Price, timing, and implementation can defeat the business thesis

PRICE

A weak company can rally

Good news or less-bad results can force the short higher. Short losses can keep growing.

TIMING

The gap may widen first

A correct long-term judgment may not pay off during the competition.

IMPLEMENTATION

Both sides must work

Check fills, financing, borrow costs, and changing exposures.

Follow both businesses, both prices, and both legs of the trade. A temporary mismatch changes the intended exposure.

How will we choose the portfolio weights?

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Connect the allocation method to the evidence your team can support

Approach	What drives the weights?	What must be assessed?
Return and risk	Expected return versus joint portfolio risk.	Return forecasts, covariance, and estimation uncertainty.
Risk budgets	A chosen distribution of measured risk.	Risk measure, covariance, and the reason for the budget.
Benchmark-relative	Deliberate departures from a reference mix.	Evidence for each departure and acceptable relative risk.
Simple rules	Equal weights or a stable strategic allocation.	Overlap, risk concentration, and rebalancing costs.

For a security, signed position weight = market value / account equity. Choosing the target is a portfolio decision.

Make the allocation problem explicit

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One example: trade expected portfolio return against variance, subject to constraints

A MEAN-VARIANCE OBJECTIVE

$$\max_w w^T \mu - \frac{\lambda}{2} w^T \Sigma w$$

EXPECTED RETURN

$w^T \mu$ combines portfolio weights and expected asset returns.

PORTFOLIO RISK

$w^T \Sigma w$ includes individual risk and co-movement.

PREFERENCE & LIMITS

$\lambda > 0$ sets the risk penalty. Feasible weights must satisfy constraints.

Estimate → solve within constraints → vary the inputs → compare with simpler allocations.

Look through the holdings to the exposures

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Different instruments can depend on the same economic outcome

Holding combination	Shared exposure	What to inspect
Broad equity + sector ETF	Overlapping companies and earnings drivers.	Underlying holdings and sector concentration.
Government + corporate bonds	Interest rates, plus additional credit exposure.	Duration, credit quality, and spreads.
Foreign assets + CAD listing	Underlying currency exposure can remain.	Fund hedging policy and reporting currency.
Industry long-short pair	Shared market and sector conditions.	Relative thesis, beta, and exposures left over.

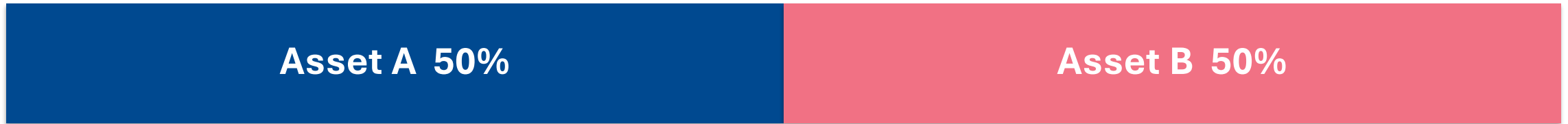
The portfolio is a combination of economic exposures, even when each security was selected for a different reason.

Half the capital can carry most of the risk

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Two uncorrelated assets: 50% weights, but 30% and 10% volatility

Capital allocation



Contribution to portfolio variance



A variance term = $(0.5 \times 0.30)^2 = 0.0225$; B = $(0.5 \times 0.10)^2 = 0.0025$. A share = $0.0225 / 0.025 = 90\%$.

How much risk does each holding contribute?

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Attribute portfolio volatility using weights and covariances

CONTRIBUTION TO PORTFOLIO VOLATILITY

$$RC_i = \frac{w_i (\Sigma w)_i}{\sigma_p}$$

w = vector of weights; Σ = covariance matrix; $(\Sigma w)_i$ = covariance of asset i with the portfolio; σ_p = portfolio volatility. RC_i is a volatility contribution.

Contributions sum to portfolio volatility. Their shares can reveal concentration, including negative contributions from offsets.

Trace a common shock through the portfolio

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Start with an economic mechanism; calibrate magnitudes before estimating losses

Shock	Transmission	Exposures to examine
Demand slowdown	Sales and earnings weaken; credit concerns rise.	Equities, corporate spreads, and government-rate sensitivity.
Persistent supply costs	Margins face pressure; inflation compensation may rise.	Equities and duration can both be vulnerable.
Funding or liquidity stress	Financing tightens; spreads widen; selling becomes costly.	Leverage, short financing, and the ability to exit positions.

Use a consistent horizon, currency, and financing treatment. Quantitative shocks need an explicit calibration basis.

Prices move the portfolio even when we do nothing

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Separate restoring a target from changing your investment view

What changed?	Meaning	Portfolio response
Only the market value	A holding grew beyond its intended weight.	Rebalance toward the existing target.
The investment evidence	Our expected payoff or risk changed.	Reconsider the target itself.
A competition constraint	An actual weight or cash level breaches a limit.	Restore compliance and check the account.

Trade the difference between the target position and the current holding. Allow for open orders, costs, and currency.

What would convince us that the idea is wrong?

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Choose evidence that can change the decision

COMPANY VIEW

Test the business mechanism

For the retailers: inventory, margins, cash generation, and what prices assume.

EVENT VIEW

Update the possible outcomes

For the takeover: approvals, financing, timing, and stand-alone failure value.

SYSTEMATIC VIEW

Test beyond the fitted history

Use information available at the time; allow for costs and unstable parameters.

A good result can come from luck. A useful investment argument can be tested against unfavorable evidence.

What does the account actually own, net of obligations?

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Read Portfolio Summary before interpreting cash or placing an order

PORTFOLIO VALUE

$$PV = \text{Long market value} + \text{Cash balance} - \text{Short market value}$$

Use all components in the account's reporting currency. Short market value is a positive liability magnitude.

Short-sale proceeds add cash and an obligation. Read the loan and signed cash fields without counting borrowing twice.

Buying power can include a loan

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Trading capacity and account wealth are different numbers

STARTING CAPACITY RELATIONSHIP

$$\text{Buying Power} = \text{Initial Cash} + \text{Margin Amount}$$

Margin Amount means the borrowing capacity included in buying power; it is not cash already earned.

Portfolio Summary shows the loan balance. Current remaining buying power changes with the account; read the live field.

The same trade earns less after financing

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Session interest rates and stock-price thresholds

MINIMUM STOCK PRICE: BUYING

3

MINIMUM STOCK PRICE: SHORTING

3

ANNUAL INTEREST EARNED ON CASH

3%

ANNUAL INTEREST CHARGED ON LOANS

8%

Price thresholds follow the ticket's quoted-price convention. Financing affects the portfolio's return.

Check the boundaries before submitting the trade

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RPMC account limits, trading period, and scoring

CAD 1m

Initial cash

20%

Maximum per security / ETF

<10%

Cash position

1 trade

Minimum every two weeks

Trading period: October 9, 2026 – March 12, 2027

Final scoring: Sharpe ranking 50% + presentation 50%*

U.S. / Canadian listings; approved universe and session settings apply.

*Presentation is for selected top teams; ranking reflects penalties.

Monitor position weights, cash, and trading activity throughout the competition.

The quantity field means different things across instruments

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Translate units into economic exposure before placing the order

Instrument	Before preview	After execution
Individual U.S. bond	Issuer; government / corporate; maturity; face-value unit; price / yield; currency	Verify face amount, transaction value, any accrued interest, and exposure.
Bond ETF	Fund identity; holdings; duration; share quantity; currency hedge	Verify shares, allocation, and underlying bond / FX exposure.
Forex	Pair direction; base / quote; quantity unit; notional; margin / financing	Verify the signed currency exposure, loan / margin fields, and hedge size.

Use the session's actual units and preview. Do not copy stock-share conventions into a bond or Forex ticket.

Delayed 15-min quote and Bid-ask spread

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The last trade, the bid, and the ask answer different questions

BID

\$49.98

A buyer's offer

ASK

\$50.02

A seller's request

Bid-ask spread: \$0.04

Delayed quote

A larger order can reach worse prices

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Follow a 600-share buy through a simplified order book

Ask price	Shares offered	Shares bought
\$50.02	100	100
\$50.05	300	300
\$50.10	600	200

$$\text{Average fill} = \frac{100 \times 50.02 + 300 \times 50.05 + 200 \times 50.10}{600} \approx 50.0617$$

Hypothetical, unchanged order book: buy 600 shares. The best displayed price may not cover the full order.

What instruction do you want the order to carry?

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Choose between urgency, a price boundary, and a trigger

Order	What it instructs	What it does not promise
Market	Trade at available market prices	A particular execution price
Limit	Buy at or below / sell at or above a limit	That the order will fill
Stop-market	Trigger a market order at a stop level	Execution at the stop price
Trailing stop	Move a stop with favorable price changes	Protection against gaps or slippage

Verify the actual order choices and trigger conventions available in your session.

Start from the position you have

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Buy, sell, short, and cover make different changes

Current position	Intention	Action to verify
Own shares	Reduce or close a positive holding	Sell
No short position	Open exposure that benefits from a decline	Short / Sell Short
Short shares	Reduce or close the short	Cover / Buy to Cover
No holding or a long	Open or add to a positive holding	Buy

The exact labels and permitted transitions are session-specific. Verify the resulting signed quantity.

Follow one trade from intention to confirmed holding

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Portfolio Summary → ticket → order status → positions and transactions

- 1 Portfolio Summary: confirm session, positions, cash, buying power, loans, and open orders.
- 2 Portfolio Simulation → Make a Trade → Stocks/ETFs: select market and instrument.
- 3 Set action, quantity, order type, and duration for the intended position change.
- 4 Preview transaction value and charges; record the investment reason; submit.
- 5 Check order status and any partial fill, rejection, or cancellation.
- 6 Verify the fill in positions and transaction history; recheck weights and cash.

Check open orders, partial fills, and cancellation requests before submitting another order.

What would a reference portfolio have earned?

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Benchmark comparison adds market context to the result

Ticker	Exposure	Weight
SPY	U.S. equities	35%
XIC.TO	Canadian equities	15%
IEMG.K	Emerging-market equities	20%
XBB.TO	Canadian bonds	20%
GLD	Gold	10%

FIND THE BENCHMARK

Portfolio Simulation



My Portfolio



Portfolio Ranking

Find: RPMC Benchmark

Compare the same dates and currency. Benchmark outperformance and Sharpe answer different questions.

Connect the idea, the portfolio, and the result

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An investment argument becomes stronger when every part fits together

INVESTMENT VIEW

Why this opportunity?

Connect the expected return to evidence and the price being paid.

PORTFOLIO ROLE

Why this exposure and weight?

Explain how the holding behaves with the rest of the account.

WHAT WE LEARNED

What drove the result?

Relate performance to decisions, market conditions, and new evidence.

Build a portfolio your team can investigate, implement, and explain.